

Chapter 5

Risk Management

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Definition of Risk:

Risk refers to the potential for events or conditions to have a negative impact on the successful completion of a project.

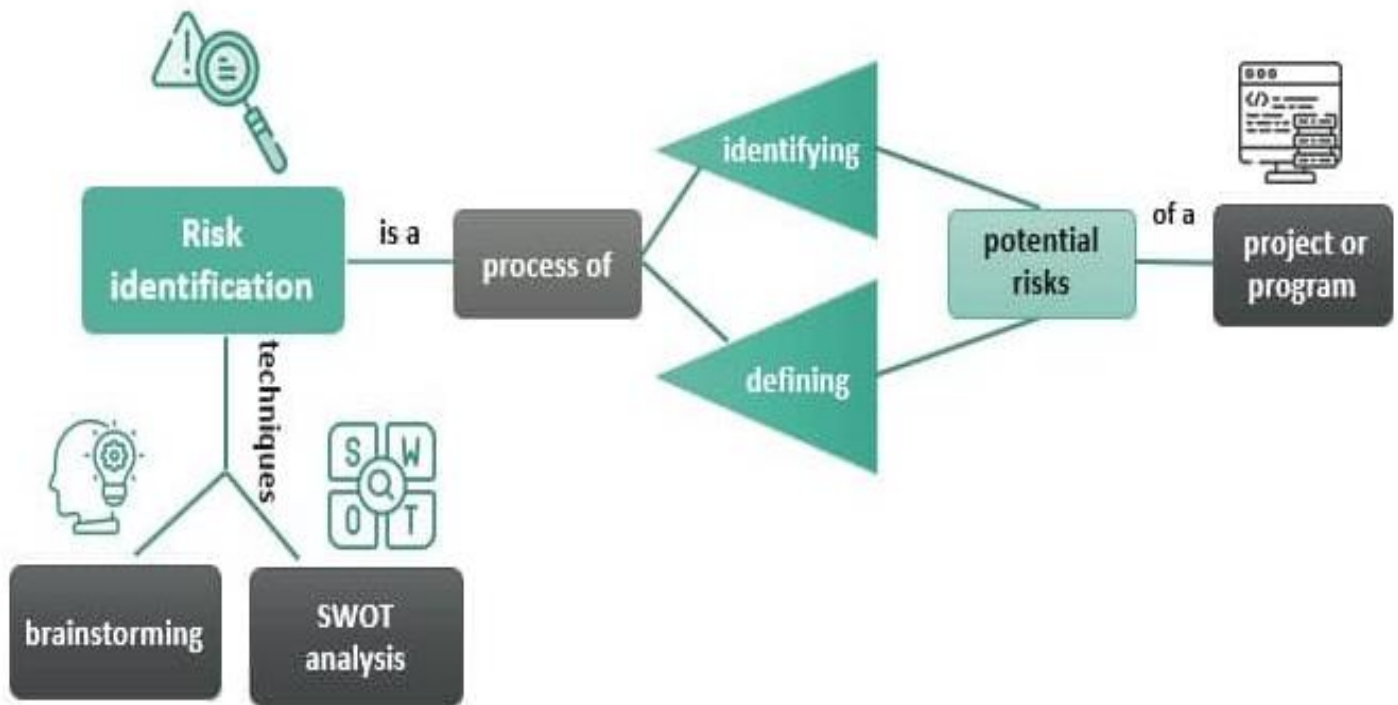
"Tomorrow problems are today's risk." Hence, a clear definition of a "risk" is a problem that could cause some loss or threaten the progress of the project, but which has not happened yet.

These potential issues might harm cost, schedule or technical success of the project and the quality of our software device, or project team morale.

Risk Identification:

What Is Risk Identification?

Risk identification is the process of identifying and defining potential risks that could impact the successful completion of a project or program.



- Risk identification is identifying and defining potential risks that could impact the success of a project.
- There are several techniques organizations can use to identify risks, including brainstorming, root cause analysis, SWOT analysis, and expert judgment.
- The risk identification process includes defining the project scope, identifying potential risks, assessing the likelihood and impact of each risk, and

developing mitigation plans for the most critical risks.

Risk Identification Process:

- 1. Identifying potential risks:** In this step, the project team and stakeholders identify potential risks that could impact the project.

This includes internal factors, such as resource shortages, and external factors, such as changes in regulations or market conditions.

- 2. Assessing the likelihood and impact:** Once potential risks have been identified, the next step is to assess the likelihood and impact of each risk.

The likelihood refers to the probability of the risk occurring, while the impact refers to the potential consequences if the risk does occur.

- 3. Prioritizing risks:** Based on the likelihood and impact of each risk, the risks should be prioritized so that the most critical risks can be addressed first.

This ensures that the project's limited resources go toward the most important risks.

4. Developing risk mitigation plans: For the most critical risks, the project team should develop risk mitigation plans to minimize the potential impact of the risks.

Hence, this may include purchasing insurance, making changes to the project schedule, or developing contingency plans.

5. Continuously monitoring and updating: Finally, the risk management plan should be continuously monitored and updated as the project progresses.

New risks may emerge, and the likelihood and impact of existing risks may change, so it is important to keep the risk management plan up-to-date.

This ensures the project gets protection against potential risks throughout its lifecycle.

Techniques for risk identification:

1. Project Documentation Review:

- Examine project documentation, including project plans, requirements specifications, design documents, and any other relevant documentation, to identify potential risks.

2. Stakeholder Interviews:

- Engage with project stakeholders, including team members, clients, and subject matter experts, to gather their insights on potential risks. Different stakeholders may have different perspectives on what could go wrong.

3. Brainstorming Sessions:

- Conduct brainstorming sessions with project team members to generate a list of potential risks. Encourage open and honest communication to capture a broad range of perspectives.

4. Checklists:

- Use risk checklists that are specific to software development to systematically identify common risks. Checklists can include categories such as technical, project management, requirements, and external factors.

5. SWOT Analysis:

- Perform a SWOT (Strengths, Weaknesses, Opportunities, Threats) analysis to identify both internal and external factors that could pose risks to the project.

Advantages of risk identification

- Early identification of potential risks.
- Improved planning and preparation for potential risks.
- A better understanding of the risks involved in a project.
- Improved communication and collaboration among stakeholders.

Disadvantages of risk identification

- Time and resource consumption.
- Difficulty in accurately predicting all risks.
- Potential for over-analyzing risks and losing focus on the project's objectives.

Risk planning:

Risk planning is the **process of identifying, prioritizing, and managing risk.**

Every project or initiative has **objectives**, that is, goals that it seeks to accomplish. These are often called Critical Success Factors (CSF).

Risk events threaten the successful completion of these critical success factors.

Thus, risk planning involves identifying the most important risk events in advance, prioritizing them, and developing the appropriate risk response plans.

There are three steps to risk planning:

1. Identifying Risks
2. Prioritizing Risks
3. Determining Response Plans

1. Identifying Risks

The first step in risk planning is to identify potential risks that could impact your project.

This involves brainstorming and considering all possible risks, both **internal and external**.

Internal risks are those that originate **within your project**, such as **resource constraints, technical challenges, or communication breakdowns**.

External risks are those that come from outside your project, such as changes in **market conditions, natural disasters, or regulatory changes**.

To effectively identify risks, consider the following questions:

- What could go wrong during the project?
- What are the potential threats to the project's objectives?
- What are the areas of uncertainty that could hinder project progress?

Use various techniques to gather information for risk identification, such as:

- Brainstorming sessions with project stakeholders
- Reviewing project documentation and historical data
- Conducting risk workshops with experts
- Analyzing similar projects and their risk experiences

2. Prioritizing Risks

Once you have identified potential risks, the next step is to prioritize them based on **their likelihood of occurrence and the potential impact** they could have on your project.

This involves assessing the severity and probability of each risk.

Severity refers to the degree to which a risk could negatively impact your project. Consider factors such as:

- Cost or financial implications of the risk
- Impact on project schedule or deadlines
- Potential damage to project quality or deliverables
- Reputation or stakeholder impact

Probability refers to the likelihood of a risk occurring. Consider factors such as:

- Historical data on similar risks
- Expert assessments and opinions
- Project complexity and uncertainty factors

Use a risk matrix or other prioritization tool to evaluate and categorize risks based on their severity and probability. This will help you focus your attention and resources on the most significant risks.

3. Determining Response Plans

For each prioritized risk, **develop a response plan that outlines how you will address the risk** if it occurs.

The goal is to either eliminate, avoid, mitigate, or transfer the risk.

- Eliminate: Remove the source of the risk altogether, if possible.
- Avoid: Change your approach to avoid the risk altogether.
- Mitigate: Reduce the likelihood or impact of the risk if it occurs.
- Transfer: Shift the risk to another party, such as through insurance.

Develop specific action plans for each response strategy, including who is responsible, what resources will be needed, and a timeline for implementation. Regularly review and update your risk response plans as your project progresses.

Risk evaluation:

Risk evaluation is a crucial step in risk management, **enabling to prioritize risks and make informed decisions about how to mitigate them.**

It involves **assessing the severity** (potential impact of a risk) and likelihood (probability) of potential risks to determine their overall impact on the objectives.

Purpose of Risk Evaluation

Risk evaluation serves several critical purposes:

1. **Assess Risk Impact:** It helps to understand the **potential impact of risks on their objectives**, enabling them to focus on the most significant ones.
2. **Prioritize Risks:** By evaluating risks, we can **prioritize them based on their severity and likelihood**, ensuring that resources are allocated effectively.
3. **Inform Decision-Making:** Risk evaluation provides **valuable insights to inform decision-making** about risk mitigation strategies and resource allocation.
4. **Communicate Risks:** It facilitates **clear and concise communication of risks to stakeholders**, fostering transparency and accountability.

Steps in Risk Evaluation

The risk evaluation process typically involves the following steps:

1. **Identify Risks:** The first step is to identify all potential risks that could impact the organization.

This can be done through brainstorming, reviewing historical data, and conducting risk workshops.

2. **Analyze Risks:** Once risks have been identified, they need to be analyzed to determine their severity and likelihood of occurrence. Severity refers to the potential impact of the risk, while likelihood refers to the probability of the risk occurring.
3. **Prioritize Risks:** Risks should be prioritized based on their severity and likelihood of occurrence. This will help the organization focus its resources on the most significant risks.
4. **Develop Mitigation Strategies:** For each prioritized risk, a mitigation strategy should be developed. This strategy should outline the steps that will be taken to reduce the likelihood or impact of the risk.
5. **Monitor and Review Risks:** Risks should be monitored on an ongoing basis to ensure that mitigation strategies are effective and that new risks are identified.

Risk Management:

risk management is the process of identifying, assessing, and prioritizing potential risks that could affect the project's success.

In other words, it is the process of making decisions about how to best deal with potential risks.

This involves considering both internal and external factors that could impact the project's objectives, timelines, and budget.



The goals of risk management are to:

1. **Increase the likelihood of project success:** By proactively identifying and addressing potential risks, projects are more likely to meet their objectives, deadlines, and budget constraints.
2. **Reduce costs and rework:** Mitigating risks upfront can prevent costly rework, delays, and resource overruns later in the project.
3. **Improve software quality and reliability:** Addressing potential technical and project management risks early on can enhance software quality and reliability.
4. **Enhance stakeholder confidence:** Effective risk management demonstrates to stakeholders that the project team is proactive and committed to delivering success.
5. **Promote continuous improvement:** Risk management provides valuable insights for continuous improvement of project management practices and risk identification capabilities.

Risk Management Process:

Effective risk management in software projects involves a structured approach to identifying, assessing, and mitigating risks throughout the project lifecycle. The following steps outline a common risk management process:

1. **Risk Identification:** Identify potential risks through brainstorming, risk workshops, historical data analysis, and expert consultations.
2. **Risk Assessment:** Assess the severity and likelihood of each identified risk using risk matrices or other assessment tools.
3. **Risk Prioritization:** Prioritize risks based on their impact on project objectives, timelines, and budget.
4. **Risk Mitigation:** Develop mitigation strategies for each prioritized risk, such as:
 - **Avoiding the Risk:** Modifying project plans or requirements to eliminate the risk altogether.
 - **Reducing Likelihood:** Implementing controls, training, or process improvements to reduce the probability of the risk occurring.

- Reducing Impact: Preparing contingency plans or fallback strategies to minimize the impact if the risk occurs.
5. Risk Monitoring and Review: Continuously monitor and review risks throughout the project, evaluating the effectiveness of mitigation strategies and identifying new risks.

Categories of Risk:

1. **Schedule Risk** : Schedule related risks **refers to time related risks or project delivery related** planning risks.

The wrong schedule affects the project development and delivery.

These risks are mainly indicates to running behind time as a result project development doesn't progress timely and it directly impacts to delivery of project.

Finally if schedule risks are not managed properly it gives rise to project failure and at last it affect to organization/company economy very badly.

Some reasons for Schedule risks –

- Time is not estimated perfectly
- Improper resource allocation
- Tracking of resources like system, skill, staff etc
- Frequent project scope expansion
- Failure in function identification and its' completion

2. Budget Risk : Budget related risks refers to the monetary risks mainly it occurs due to budget overruns.

Always the financial aspect for the project should be managed as per decided but if financial aspect of project mismanaged then there budget concerns will arise by giving rise to budget risks.

So proper finance distribution and management are required for the success of project otherwise it may lead to project failure.

Some reasons for Budget risks –

- Wrong/Improper budget estimation

- Unexpected Project Scope expansion
- Mismanagement in budget handling
- Cost overruns
- Improper tracking of Budget

3. Operational Risks : Operational risk refers to the procedural risks means these are the risks which happen in day-to-day operational activities during project development due to improper process implementation or some external operational risks. Some reasons for Operational risks –

- Insufficient resources
- Conflict between tasks and employees
- Improper management of tasks
- No proper planning about project
- Less number of skilled people
- Lack of communication and cooperation
- Lack of clarity in roles and responsibilities
- Insufficient training

4. Technical Risks : Technical risks refers to the **functional risk or performance risk** which means this technical risk mainly associated with functionality of product or performance part of the software product. Some reasons for Technical risks –

- Frequent changes in requirement
- Less use of future technologies
- Less number of skilled employee
- High complexity in implementation
- Improper integration of modules

5. Programmatic Risks : Programmatic risks refers to the **external risk or other unavoidable risks**.

These are the external risks which are unavoidable in nature.

These risks come from outside and it is out of control of programs.

Some reasons for Programmatic risks –

- Rapid development of market
- Running out of fund / Limited fund for project development

- Changes in Government rules/policy
- Loss of contracts due to any reason

Framework for dealing with risk:

Dealing with risk in software project management is a critical aspect of ensuring project success.

Implementing a framework to manage risks helps in identifying potential issues early, developing strategies to mitigate them, and ultimately minimizing the impact on the project.

Framework for dealing with risk in software project management:

1. Plan risk management

- Establish risk management procedures: Define the process for identifying, analyzing, prioritizing, and managing risks throughout the project lifecycle.
- Assign risk management roles and responsibilities: Clarify who is accountable for each aspect of risk management, ensuring clear ownership and accountability.

- Allocate resources for risk management: Set aside sufficient time, budget, and resources to effectively manage project risks.

2. Identify risks

- Brainstorm potential risks: Gather stakeholders and team members to brainstorm potential risks that could impact the project's objectives.
- Use risk identification techniques: Utilize risk identification techniques such as interviews, questionnaires, and historical data analysis to uncover potential risks.
- Categorize identified risks: Group identified risks into categories based on their nature, such as technical, schedule, cost, or resource-related risks.

3. Analyze and prioritize risks

- Assess likelihood and impact: Evaluate the likelihood and potential impact of each identified risk to determine its overall severity.
- Prioritize risks: Prioritize risks based on their severity, considering both the likelihood of occurrence and the potential impact on the project.

- Document risks: Document all identified risks, their likelihood, impact, and priority in a risk register or similar tool.

4. Develop risk mitigation plans

- Develop mitigation strategies: For each prioritized risk, develop a mitigation strategy to reduce its likelihood or impact.
- Assign mitigation responsibilities: Assign ownership and accountability for implementing and monitoring risk mitigation strategies.
- Estimate mitigation costs: Estimate the resources and costs associated with implementing each risk mitigation strategy.

5. Monitor and control risks

- Track risk progress: Regularly monitor the status of identified risks and the effectiveness of mitigation strategies.
- Update risk register: Update the risk register as risk status or mitigation strategies change.
- Communicate risk information: Communicate risk information to stakeholders regularly to keep them informed about potential issues and mitigation measures.

6. Continuously improve risk management

- Review risk management process: Regularly review and evaluate the effectiveness of the risk management process and make improvements as needed.
- Learn from experiences: Reflect on past project risks and mitigation strategies to continuously improve risk management practices.
- Adapt to changing circumstances: Adjust risk management plans and strategies as project conditions and external factors evolve.

Evaluating risk to Schedule:

Evaluating risk to the schedule is a crucial aspect of software project management, as it **helps project managers identify potential delays** and proactively take measures to mitigate them.

Effective schedule risk evaluation involves **assessing the likelihood and impact of potential risks** and **developing contingency plans to address them**.

Assessing Likelihood and Impact

To evaluate schedule risk, project managers should consider two key factors:

1. **Likelihood:** The probability that a risk will occur. This assessment involves considering historical data, expert opinions, and project-specific factors that could influence the risk's occurrence.
2. **Impact:** The severity of the consequences if the risk materializes. This assessment involves evaluating the potential delay to the project schedule, the impact on project costs and resources, and the overall impact on project success.

Prioritizing Schedule Risks

After assessing the likelihood and impact of each risk, project managers should prioritize them based on their overall severity. This prioritization helps focus resources on the most critical risks and ensures that mitigation efforts are directed towards the areas of greatest potential impact.

Developing Contingency Plans

For each prioritized schedule risk, project managers should develop contingency plans to address the risk if it occurs. These plans should outline specific actions and strategies

to mitigate the impact of the risk and minimize the delay to the project schedule.

Strategies for Schedule Risk Evaluation

Several quantitative and qualitative techniques can be used to evaluate schedule risk, including:

1. Delphi method: A structured process for gathering and refining expert opinions on potential risks and their likelihood and impact.
2. Monte Carlo simulation: A statistical method that generates multiple scenarios based on the likelihood and impact of various risks, providing a probabilistic assessment of project schedule outcomes.
3. PERT (Program Evaluation and Review Technique): A method for estimating project task durations and identifying potential schedule bottlenecks.
4. Earned Value Management (EVM): A technique for comparing planned schedule progress to actual progress, identifying potential schedule variances early on.
5. Risk matrices: A visual tool for categorizing risks based on their likelihood and impact, facilitating prioritization and decision-making.

Continuous Monitoring and Improvement

Schedule risk evaluation is an ongoing process that should be revisited throughout the project lifecycle. As project conditions change and new information becomes available, project managers should reassess the likelihood and impact of identified risks and adjust mitigation strategies accordingly.